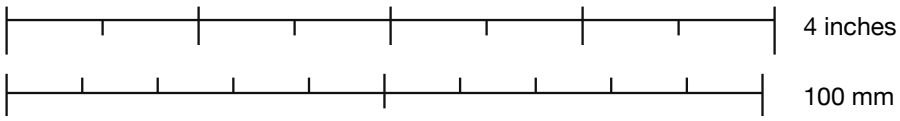
Jacques O'Lantern: carving template

For the 13" hollow plastic craft pumpkin. Every opening is sized for the parts that go behind it: Logitech C920 webcam, one Logitech Z150 speaker, LED or NeoPixel Jewel eyes.

> **Print at 100% ("Actual size"), not "Fit to page".** Then check the ruler below: it must measure exactly 4 in and 100 mm. If it doesn't, the cuts will be the wrong size.



What you're cutting

Opening	Size	Where	Why that size
Eyes (2)	Round, 1.5" across, centres 3" apart	Eye centres 1.4" above the pumpkin's widest line	A NeoPixel Jewel (0.9") fits behind each with room for a paper diffuser. Round eyes suit Jacques's spectacles.
Mouth	Grin 6" wide, 2" tall at the centre, corners turned up	Top of the mouth 0.3" below the widest line	Clears the webcam's 78° view and both of its mics, and lets the speaker's sound out.
Back hatch	5" × 7", rounded corners, plus a cable notch	Centred on the back, bottom edge about 1.5" above the base	One Z150 speaker (about 4.5" × 6.5") goes in through it. Keep the cut-out as a door.

Before you cut

1. **Find the widest line.** Wrap a strip of masking tape around the pumpkin at its widest point. That's the reference line marked on the template.
2. **Check Jacques's spectacles.** If you'll hang real costume glasses over the eyes, measure between the centres of their lenses. If it isn't about 3", move the eye centres on the template to match before you cut.
3. **Pick the face.** Choose the smoothest side of the pumpkin, and put the back hatch on the opposite side.

Cutting

1. Cut out each template along the outer dashed border. Snip the short **slit marks** on the edges so the paper bends around the curve.
2. Tape the face template on: the **centre line vertical**, and the template's **widest-line marker on your tape line**. Trace every solid line with a fine marker, and mark the camera crosshair.
3. Make a **starter hole** at each black dot (drill, or melt with the iron). Then cut from hole to hole. Forcing a cold blade into thin plastic cracks it.
4. **Hot knife** (soldering iron with a spare tip): **outdoors or with a fan**, because melting plastic gives off nasty fumes. Or use craft-pumpkin carving saws.
5. Do the back hatch the same way, but keep the cut-out piece in one piece for the door.

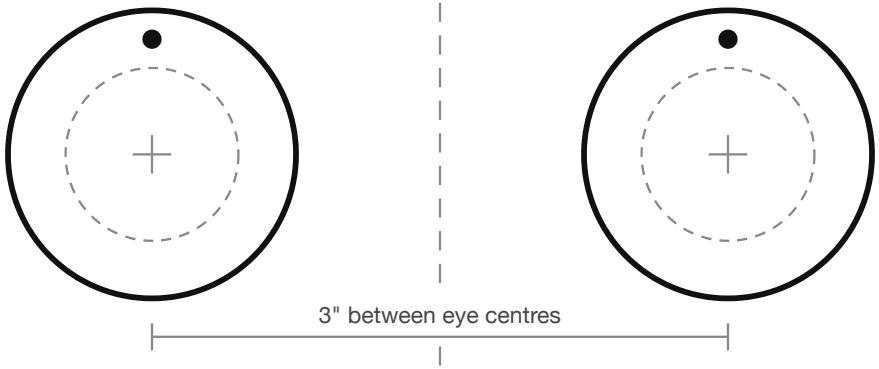
Fitting the parts

- **Mouth:** hot-glue black tulle inside it with 0.5" of overlap all round. From outside the mouth looks dark; the camera sees through and sound passes through.
- **Webcam:** lens on the crosshair, touching the tulle. Hold it with its clip or a hot-glued foam block. The template shows where its body sits.
- **Aim:** stand the pumpkin so the mouth is about 26–30" off the porch floor (a crate helps). Aim level to about 10° upward, then check on the dashboard: a child at the candy bowl should be in frame from the waist up, with an adult's head in frame too.
- **Speaker:** in through the hatch, facing the mouth. Afterwards, re-check echo: stay quiet while Jacques talks, and make sure no 👩 lines echo his words.
- **Eyes:** LED (or Jewel) centred behind each eye with white paper or tulle in front. Make the mount easy to open, because the Jewels replace the LEDs later.

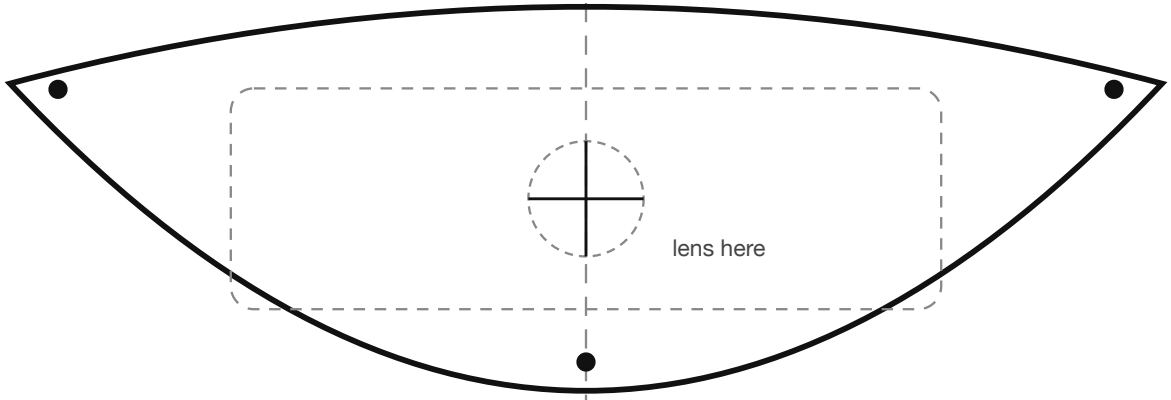
slit

FRONT: face template (actual size)

Solid lines = cut. Dashed = guides, don't cut. Black dots = starter holes.



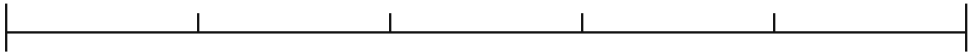
widest line of the pumpkin



Mouth: 6" wide, 2" tall at the centre

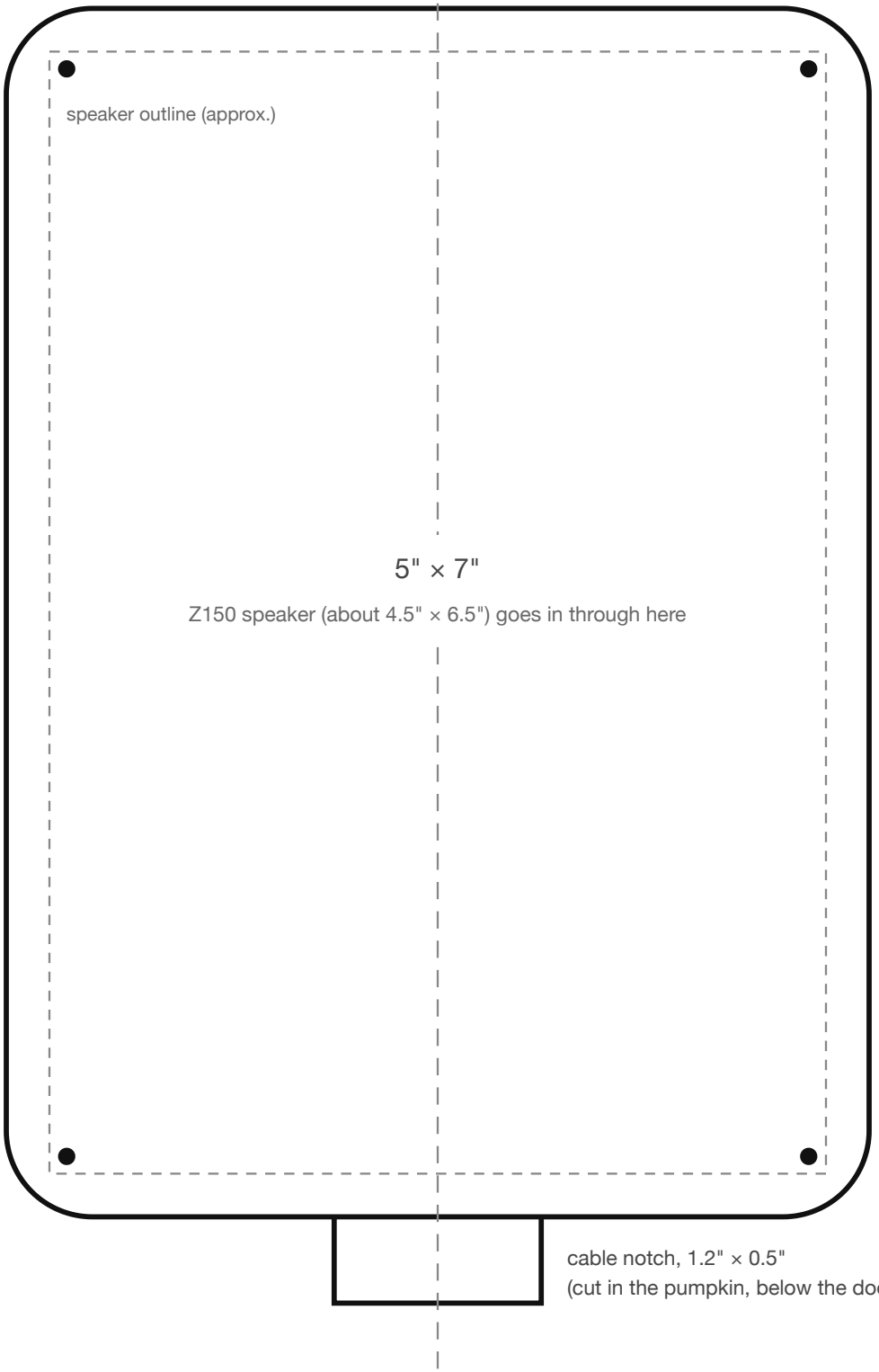
centre line (keep vertical)

Check this ruler measures exactly 5" before cutting



BACK: service hatch (actual size)

Solid lines = cut. Keep the cut-out in one piece: it's the door.



Door: tape hinge along the top edge on the inside; small Velcro tabs at the bottom corners hold it shut.